

Rahul Jana

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Machine Learning Engineer with 2.5+ years building and deploying end-to-end ML and computer vision systems on AWS. Designed scalable MLOps pipelines with 98% uptime, optimized CV models(eg YOLO, UNet) for production, and contributed to open-source research with published work. Finalist in a \$1.4M global AI innovation challenge.

Experience

TIF- AWADH, Indian Institute of Technology - Ropar

Software Engineer

April 2023 - Present

- Architected and maintained AWS-based ML deployment pipelines with **98% uptime** using Docker, SageMaker, and CI/CD workflows.
- Developed an accelerometer-based cattle activity detection model achieving **93% accuracy** (*published on arXiv*).
- Built a YOLOv5-based insect detection pipeline reaching **92% accuracy** for agri-automation.
- Engineered a custom cattle activity model with YOLOv5, achieving **80% accuracy**, optimizing on-device inference.
- **Reduced dataset creation time by 30%** by building a high-throughput annotation pipeline using Python & OpenCV.
- Built ML models achieving **90% accuracy** for post-harvest crop quality assessment using spectral time-series.
- Developed a U-Net + VGG16 model for room segmentation and style-transfer-based interior design automation.

Springworks

Remote

Machine Learning Intern

September 2021 - July 2022

- Built ML components for document quality detection across Govt. ID formats (Aadhar, PAN, Passport).
- Improved runtime performance of a crypto trading algorithm by **~20%** using live trading scenario.
- Awarded Intern of the Month (May 2022).

Open source Contributions, Projects & Hackathons

Deep Forest Sciences

- Finalist in the Standard Industries Chemical Innovation Challenge (\$1.4M prize pool).
- Contributed backend improvements and edge-case handling for **DeepRetro, a Retrosynthetic Pathway Discovery using Iterative LLM Reasoning [GitHub]** (*published on arXiv*).
- Improved experiment tracking, data storage, and result reuse in the **Prithvi** platform.

Skills

ML and DL: • Python • GenAI • PyTorch • FastAPI • TensorFlow • LLMs • SageMaker

Cloud and MLOps: • Linux • Docker • AWS • GitHub Actions • PostgreSQL • MongoDB • LangChain

Education

Chandigarh University

Punjab, India

CGPA: 7.35 / 10.0

- Developed a Voice-Controlled Switch Box for **Socially Relevant Projects** and **Abdul Kalam Innovation Conclave**.